GOVT 6019: Quantifying Omitted Variable Bias

Fall 2026

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Setting up the problem

- In this handout we measure omitted variable bias with algebra, for a regression with two explanatory variables.
- We want the effect of X_1 on Y, but a second variable, X_2, may also affect Y. We take the long regression (i.e., the "true" population regression function) to be

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X_1, X_2] = 0.$$

- Suppose X_2 is omitted, for example because X_2 is not in our data. Then we can only fit the short regression

$$Y = \beta_0^* + \beta_1^* X_1 + v.$$

  Call its estimated slope $\hat{\beta}_1^*$.
- Our target is the "true" β_1. With the short regression we can only estimate a different quantity, β_1^*. We want to know how far apart β_1^* and β_1 are.
- The **bias** is the difference between the short-regression slope β_1^* and the "true" β_1 from the long regression:

$$\text{Bias} = \beta_1^* - \beta_1.$$

- When we run a study or read one, we want to know how large this bias is and whether the bias is positive or negative. Below we derive a formula for the bias.

Deriving a term for the bias expression

We rewrite the bias in terms of quantities we may know something about, in six steps.

Step 1. Write the regression of X_2 on X_1. *"auxiliary regression"*

$$X_2 = \delta_0 + \delta_1 X_1 + w, \qquad \mathbb{E}[w \mid X_1] = 0$$

δ_1 is the slope of the relationship between X_1 and X_2. *i.e., if we had X_2, what would be the relationship between X_2 and X_1?*

Step 2. In the long regression, replace X_2 with the right-hand side of the Step 1 equation. *(i.e., what X_2 is in terms of X_1)*

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$
$$Y = \beta_0 + \beta_1 X_1 + \beta_2(\delta_0 + \delta_1 X_1 + w) + \varepsilon$$

X_2 no longer appears in the equation.

Step 3. Multiply β_2 into each term in the parentheses.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2\delta_0 + \beta_2\delta_1 X_1 + \beta_2 w + \varepsilon$$

Step 4. Collect the constant terms, the terms in X_1 and the error terms.

$$Y = [\beta_0 + \beta_2\delta_0] + [\beta_1 + \beta_2\delta_1]\, X_1 + [\beta_2 w + \varepsilon]$$

Step 5. Each bracket in the Step 4 equation matches one term of the short regression:

$$Y = \underbrace{[\beta_0 + \beta_2\delta_0]}_{\beta_0^*} + \underbrace{[\beta_1 + \beta_2\delta_1]}_{\beta_1^*}\, X_1 + \underbrace{[\beta_2 w + \varepsilon]}_{v}$$

The error $\beta_2 w + \varepsilon$ has mean 0 at every value of X_1, because w and ε both do. For ε, $\mathbb{E}[\varepsilon \mid X_1] = 0$ follows from $\mathbb{E}[\varepsilon \mid X_1, X_2] = 0$ by the law of iterated expectations. So the Step 4 equation is the short regression. Its slope is

$$\beta_1^* = \beta_1 + \underbrace{\beta_2\delta_1}_{\text{“junk”}}.$$

- β_1^*: *coefficient in the regression where we didn't include X_2 (regression of Y on X_1)*

Step 6. Subtract β_1 from both sides. The result is the formula for omitted variable bias:

$$\boxed{\text{Bias} = \beta_1^* - \beta_1 = \beta_2\,\delta_1}$$

- β_2: *actual effect of X_2 on Y*
- δ_1: *are X_1 and X_2 correlated with each other? (we can't estimate δ_1 without X_2)*

So $\mathbb{E}[\hat{\beta}_1^*] = \beta_1^*$: on average $\hat{\beta}_1^*$ misses β_1 by $\beta_2\delta_1$.

When is there bias?

- The **size of the bias** is a function of the sizes of β_2 and δ_1. The two conditions for omitted variable bias come from that product. $\hat{\beta}_1^*$ is biased only if both hold:
  1. X_2 is a “true” determinant of Y, i.e., $\beta_2 \neq 0$; and
  2. X_2 is correlated with X_1, i.e., $\delta_1 \neq 0$.
- **If both conditions hold**, $\hat{\beta}_1^*$ is a biased estimator of β_1.
- If $\delta_1 = 0$ or $\beta_2 = 0$, the bias equals zero. In words:
  - If $\beta_2 = 0$, X_2 does not affect Y, so leaving X_2 out omits nothing.
  - If $\delta_1 = 0$, X_1 and X_2 are uncorrelated. The short-regression slope on X_1 then includes no part of the effect of X_2.
- *If you know (or can guess) the signs of β_2 and δ_1, even if you don't have X_2, you can say something about the direction of the bias.*

Can we compute the bias exactly?

- Usually we cannot use the formula to compute the exact bias in our short-regression estimate. To compute $\beta_2\delta_1$ we need estimates of β_2 and δ_1. Both estimates require data on X_2. X_2 is the variable we do not have.

The sign of the bias

- The exact size of the bias is often hard to compute. The **sign of the bias** is often easier to work out. The sign of the bias is the sign of $\beta_2\delta_1$. We can often make an educated guess about the signs of β_2 and δ_1, even when we know little about their sizes.
- δ_1 has the same sign as the correlation between X_1 and X_2. So the sign of the bias is a function of the signs of $\text{Corr}(X_1, X_2)$ and β_2.
- In the table, $+$ means positive bias and $-$ means negative bias.

	$\text{Corr}(X_1, X_2) > 0$	$\text{Corr}(X_1, X_2) < 0$
$\beta_2 > 0$	$+$	$-$
$\beta_2 < 0$	$-$	$+$

- If the bias has the same sign as β_1, β_1^* is further from zero than β_1. If the bias has the opposite sign and is smaller in absolute value than β_1, β_1^* is closer to zero than β_1. If the bias has the opposite sign and is larger in absolute value than β_1, β_1^* has the opposite sign from β_1.